INTRODUCTION TO COMPLEXITY THEORY

Complexity theory is the study of the resources required by algorithms. The goal of complexity theory is to determine the amount of computational resources needed to perform certain computational tasks. The efforts to achieve these goals have only been partly successful.

❖ TRACTABLE AND INTERACTABLE PROBLEMS

1.TRACTABLE / EASY PROBLEMS

- The Problem that is solvable by a polynomial time algorithm is tractable.
- The upper bound is polynomial.
- There is an efficient algorithm to solve it in polynomial time.
- Polynomial time => O(n^k),for some constant k
- Time taken by an algorithm which are solvable can be O(n^k),which is very much polynomial for some constant k
- Examples for tractable problems
- ➢ Searching an unordered list
- ➢ Searching an ordered list
- ➢ Sorting a list
- ➢ Multiplication of integers
- ➢ Finding a minimum spanning tree in a graph

2.INTERACTABLE / HARD PROBLEMS

- The problem that can not be solvable by a polynomial time algorithm.
- The lower bound is exponential.
- Unable to solve them in a reasonable time.There is no efficient algorithm to solve it in polynomial time
- Exponential time => O(k^n)
- Examples of interactable problems
- ➢ Tower of Hanoi
- ➢ TSP
- ➢ Sudoku
- ➢ 0/1 knapsack
- ➢ Graph coloring

Exponential function vs polynomial

- The function p(x)=x^3 is a **polynomial**
- Here the “variale” , x ,is being raised to some constant power
- The function f(x)=x^3 is an **exponential** function
- The variable is the exponent
- **Exponential functions**
- Example: O(2^n) , O(n!) , O(n^n)

Polynomial functions:

- Any function that is O(n^k) , ie.

  Bounded from above by n^k for some constant k.

- Example: O(1) , O(log n) ,O(n log n),O(n^2),O(n^3)
- ‘Polynomial‘ is function of the form

  $a_k n^k + a_{k-1} n^{k-1} + \ldots + a_1 n + a_0$

Constant	O(1)
Logarithmic	O(logn)
Linear	O(n)
n-log-n	O(n*logn)
quadratic	O(n^2)
cubic	O(n^3)
Exponential	O(k^n)
Factorial	O(n!)
Super-exponential	O(n^n)

Deterministic algorithms

- Algorithm with the property that result of every operation is uniquely defined
- Predictable in terms of output for a certain input

Nondeterministic algorithms

- Result of every operation is not uniquely defined
- They are allowed to certain operations whose outcomes are limited to a given set of possibilities

Many problems are solved where the objective is to maximize or minimize some values, whereas in other problems we try to find whether there is a solution or not.

Hence,the problems can be categorized as follows:

Optimization problems

- shortest An optimization problem is one which asks, “What is the optimal solution to problem X?
- An optimization problem tries to find an optimal solution
- Result is a number representing an objective value
- Optimization problems are those for which the objective is to maximize or minimize some values.
- For examble, Finding the path between two vertices in a graph.

Decision problems

- A decision problem is one with yes/no answer ie, a problem if its output is a simple “no” or “yes”

UNIVERSITY QUESTIONS

- What is complexity theory?
- What Is the goal of complexity theory?
- What are tractable and interactable problems?